

Course Outline

WEB 230 JAVASCRIPT I

Course ID: 023733**Academic Year:** 2024/25**Course Description:**

This course introduces the student to the creation of dynamic Web pages using the popular Web scripting language, JavaScript. This course is perfect for beginning programmers with prior knowledge of HTML. Students will be introduced to Object-Oriented JavaScript and the debugging of JavaScript code.

Pre-Requisites:	WEB 110 AND (MIT 153 OR MAD 102 OR WEB 101 OR MIT 155)	Category:	Vocational
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Co-Requisites:	None	Course Credits:	4.00
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Special Conditions:	None	Academic Level:	Credit (Post Sec)
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Instructional Hours:	Classroom Instruction	30
	Laboratory/Workshops	30
	Other	0
	Total Hours	60

Academic Department:

Windsor:	Zekelman School of Information Technology
Chair:	Ruth Susannah
Chatham:	
Chair:	

Revised By:	M. Haug
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Last Revision:	2024/12/20
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Required Tools, Equipment, and Learning Resources:

Eloquent JavaScript, 3rd edition

by Marijn Haverbeke

Available free online at: <https://eloquentjavascript.net/>

Essential Employability Skills (EES):

Description	Teach	Assess
1) Communication: Communicate clearly, concisely and correctly in the written, spoken, and visual form that fulfills the purpose and meets the needs of the audience.		
2) Communication: Respond to written, spoken, or visual messages in a manner that ensures effective communication		
3) Numeracy: Execute mathematical operations accurately		
4) Critical Thinking: Apply a systematic approach to solve problems	✓	✓
5) Critical Thinking: Use a variety of thinking skills to anticipate and solve problems		
6) Information Management: Locate, select, organize, and document information using appropriate technology and information systems	✓	✓
7) Information Management: Analyze, evaluate and apply relevant information from a variety of sources		
8) Interpersonal: Show respect for the diverse opinions, values, belief systems, and contributions of others		
9) Interpersonal: Interact with others in groups or teams in ways that contribute to effective working relationships and the achievement of goals		
10) Personal: Manage the use of time and other resources to complete projects		
11) Personal: Take responsibility for one's own actions, decision and consequences	✓	✓

Course Learning Outcomes (CLO):

Upon successful completion of this course, the student will be able to:
(EKS = Embedded Knowledge and Skills)

- Apply industry standards in developing web sites (CLO #1)
 EKS:
 - Demonstrate the role of the Behaviour layer in the Progressive Enhancement model of Web development
 - Apply HTML5 standards using W3C recommendations in developing web sites

- Familiarize students with web accessibility standards such as the Web Content Accessibility Guidelines (WCAG).
- Apply a systematic approach to solve problems (CLO #2)
EKS:
 - Alter images based on user actions such as a mouse hover
 - Pre-load images to improve performance
 - Use array functions to manipulate groups of data
 - Use Regex to filter data
- Write code that adds appropriate behaviour to elements of a web page (CLO #3)
EKS:
 - Create and work with date objects to perform time and date operations and to display time and dates
 - Use JavaScript and CSS to change the styles of an individual or group of elements
 - Access and alter CSS attributes by accessing the Attribute elements of the DOM tree
 - Determine the characteristics of the browser window and to adjust the user interface to suit the device
 - Create JavaScript code that will validate the data entered into a form by a user
 - Extract data from a web form and to insert calculated data into elements of a web form
 - Provide feedback to a use using both popup and inline messages
- Apply dynamic behaviour to existing web pages (CLO #4)
EKS:
 - Create user interaction by attaching event handlers to DOM elements
 - Use timers to create animated effects such as automated slide shows
 - Demonstrate how to dynamically create, modify, and remove DOM elements using JavaScript
 - Analyze how to perform client-side form validation using JavaScript and regex to provide immediate feedback to users and improve the user experience.

Teaching/Learning Activities:

Activities may include, but are not limited to:

Active Learning

Labs/Computer Labs

Lecture

Assessment:**Standard/Traditional Delivery**

• Assignments	30.00%
<i>Frequency:</i> 6-10 SETS of Assignments	
<i>Description:</i> Introduce students to JavaScript programming language used to enhance the appearance and functionality of web pages through a series of hands-on assignments.	
<i>Outcomes Assessed:</i> 1, 2, 3, 4	
<i>EES Assessed:</i> 4, 6, 11	
• Final Exam	25.00%
<i>Frequency:</i> 1	
<i>Description:</i> Covers Week 1-14 materials covered in the beginning of the semester, as well as working with the DOM, advanced application of CSS, adding dynamic behavior, and JavaScript run time environments	
<i>Outcomes Assessed:</i> 1, 2, 3, 4	
<i>EES Assessed:</i> 4, 6	
• Mid term Exam	25.00%
<i>Frequency:</i> 1	
<i>Description:</i> Covers Week 1-6 materials including application of basic JavaScript syntax, proper use of industry standard coding techniques, error detection, proper use of objects and prototypes, and regex	
<i>Outcomes Assessed:</i> 1, 2, 3	
<i>EES Assessed:</i> 4, 6, 11	
• Topic Quizzes	20.00%
<i>Frequency:</i> 6-10	
<i>Description:</i> Self-test online quizzes used to progressively evaluate the level of understanding of each topic covered in the course.	
<i>Outcomes Assessed:</i> 1, 2, 3, 4	
<i>EES Assessed:</i> 6	
100%	

Note: The assessment listed in this outline represents the planned assessment method for this course. Unanticipated conditions during the delivery of the course may necessitate changes to the planned assessment. Students will receive reasonable advance notice should any changes be necessary.

Grading:

A = 80% - 100.0%
B = 70% - 79%
C = 60% - 69%
D = 50% - 59%
F = 0% - 49%

Course Content:

See EKS statements.

"Academic misconduct, including cheating of any form, will not be tolerated. Consequences may include, but are not limited to, a warning, a grade of "0" on the assignment/test/examination, or a failing grade in the course."

(Code of Students Rights and Responsibilities: Section 7.1.6)

All students and employees of this College have a right to study and work in an environment that is free from harassment and discrimination.

Accommodation Statement

The College will provide supports and services to all students with disabilities, both temporary and permanent, with valid supporting documentation. Interim accommodation requests will be received in good faith and can be provided pending receipt of medical documentation. Retroactive accommodations will be considered based on the unique circumstances of the individual matter. The College will give all Human Rights Code-related requests for accommodation meaningful consideration.

Procedure: The student is responsible to meet with a counsellor in Accessibility Services to discuss their functional limitations and accommodation needs and provide Accessibility Services with supporting documentation. Students are not required under the Ontario Human Rights Code to disclose their disability diagnosis (with the exception of Learning Disabilities) to receive accessibility supports and services and/or academic accommodations. Students are encouraged to meet with a counsellor prior to the start of a semester to provide information and arrange accommodations.